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***TrLab* User guide**

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1 Introduction

Introduction

Ontologizer [1]

Neurocomputing [10]

The Scientific World Journal [7]

Bibm 14 [6]

Evolutionary Bioinformatics [8]

Hais 16 [9]

Entropy [11]

2 Installation

Unzip *TrLab.zip* file and locate the *TrLab* folder in the path that you prefer taking into account that all applications have to be launch by command line therefore this root path will have to be referenced in the executed command (see sections 4.1 and 4.2). In the other hand, you could use the operating system tools in order to automatize this matter (e. g. *sh* files of *Unix* systems).

2.1 TrLab folder content

This is the content of *TrLab* folder:

- *TriGenApp.jar*: The *TriGen* algorithm application. With this programme you will be able to launch experiments with an wide collection of datasets. (section 4.1)
- *LabApp.jar*: The laboratory application. You will be able to analyse the *TriGen* results both individually and comparing a collection of results (section 4.2).
- *Ontologizer.jar*: The *Gene Ontology* application in order to get the *GO* analysis of *TriGen* results (<http://ontologizer.de>).
- *resources* folder: The datasets, terms files and association files required to run the applications.

2.1.1 resources folder

As you can see in Table 1, the *resources* folder has several items.

- *resources.xml*: It is the resources descriptor and manage the information of all the datasets of the system (see 2.1.2). If you want include a new dataset in the system in order to experiment with it, you have to follow several steps and alter this file (see 2.3).
- *associations* folder: The associations files in order to perform *GO* analysis.
- *terms* folder: The terms files in order to perform *GO* analysis.
- *0001* to *0014* folders: Each numbered folder has the resources of a dataset. This resources includes the gene tags, the sample tags, the time tags and one *csv* file of values for each time point. You can see the detailed content of all folders in the appendix A.

2.1.2 resources.xml format

An xml entry example

```
<dataset  
id="0001" name = "yeast" type="b" geneSize = "6178"  
sampleSize = "4" timeSize = "14" minG = "15"
```

Name	Description
<i>resources.xml</i>	resources descriptor
<i>associations</i> folder	GO associations files
<i>terms</i> folder	GO terms files
<i>0001</i> folder	yeast resources
<i>0002</i> folder	elutriation resources
<i>0003</i> folder	gds4510 resources
<i>0004</i> folder	gds4472 resources
<i>0005</i> folder	gds4262 resources
<i>0006</i> folder	gds4442 resources
<i>0007</i> folder	gse4324 resources
<i>0008</i> folder	earthquake resources
<i>0009</i> folder	synth01 resources
<i>0010</i> folder	synth02 resources
<i>0011</i> folder	synth03 resources
<i>0012</i> folder	synth04 resources
<i>0013</i> folder	synth05 resources
<i>0014</i> folder	toy01 resources

Table 1: *resources* folder content

```
maxG = "200" minC = "2" maxC = "4" minT = "3" maxT = "8"
organism = "Yeast"
description = "Complete Saccharomyces Cerevisiae experiment">
```

```
<resources separator = ";">
  <resource>yeast_t0.csv</resource>
  <resource>yeast_t1.csv</resource>
  <resource>yeast_t2.csv</resource>
  <resource>yeast_t3.csv</resource>
  <resource>yeast_t4.csv</resource>
  <resource>yeast_t5.csv</resource>
  <resource>yeast_t6.csv</resource>
  <resource>yeast_t7.csv</resource>
  <resource>yeast_t8.csv</resource>
  <resource>yeast_t9.csv</resource>
  <resource>yeast_t10.csv</resource>
  <resource>yeast_t11.csv</resource>
  <resource>yeast_t12.csv</resource>
  <resource>yeast_t13.csv</resource>
</resources>

<genes>yeast_genes.txt</genes>
<samples>yeast_samples.txt</samples>
<times>yeast_times.txt</times>

</dataset>
```

- id
- name

- type
- geneSize
- sampleSize
- timeSize
- minG
- maxG
- minC
- maxC
- minT
- maxT
- organism
- description

2.2 Available datasets

Name	Full name	Genes	Conditions	Times	Reference
yeast	<i>Yeast cell cycle</i>	6179	4	14	[14]
elutriation	<i>Elutriation</i>	7744	13	14	[14]
gds4510	<i>Mouse GDS4510</i>	22690	8	4	[5]
gds4472	<i>Human GDS4472</i>	54675	4	6	[2]
gds4262	<i>Human GDS4262</i>	54675	12	4	[12]
gds4442	<i>Mouse GDS4442</i>	45101	6	3	[4]
gse4324	<i>Mouse GSE4324</i>	22690	12	4	[3]
earthquake	<i>Earthquake</i>	90	60	7	[13]
synth01	<i>Synthetic 01</i>	4000	30	20	
synth02	<i>Synthetic 02</i>	5000	12	14	
synth03	<i>Synthetic 03</i>	2000	5	10	
synth04	<i>Synthetic 04</i>	2000	5	10	
synth05	<i>Synthetic 05</i>	2000	5	10	
toy01	<i>Toy 01</i>	5000	12	14	

Table 2: Available datasets

2.3 Including a new dataset

A new dataset entry

<dataset

```
id="" name="" type="" geneSize=""
sampleSize="" timeSize=""
minG="" maxG="" minC=""
```

```
maxC = "" minT = "" maxT = ""  
organism = "" description = "">
```

```
<resources separator = "">  
<resource>...</resource>  
</resources>  
<genes>...</genes>  
<samples>...</samples>  
<times>...</times>
```

```
</dataset>
```


3 GUI

4 Command line

4.1 *TriGenApp* commands

Trigen command

```
java -jar /path/to/folder/TrLab/TriGenApp.jar /path/to/  
configuration_file.tricfg
```

4.1.1 Input configuration file format

4.1.2 Output solution file format

4.1.3 Output files returned

4.2 *LabApp* commands

Single mode

```
java -jar /path/to/folder/TrLab/LabApp.jar  
-i /path/to/solution_file.sol  
[-a] [-f] [-o pathToOutput]
```

Multiple mode

```
java -jar /path/to/folder/TrLab/LabApp.jar  
-l /path/to/list_file.txt  
[-b, -c, -s]  
[-f] [-o pathToOutput]
```

Options:

- -i: Single input. The input is a single experiment.
- -l: Multiple analysis. The input is a set of experiments.
- -b: All the input experiments will have a biological analysis.
- -c: All the input experiments will have a common analysis.
- -s: All the input experiments will have a synthetic analysis.
- -f: Files option. It build all the results files.
- -a: Analysis option. It performs the analysis of the input.
- -o pathToOutput: It points out where the output files will be placed (complete path and file name). By default, the output files will be place in the same path that the input file.

4.2.1 Input list-file format

4.2.2 Output files returned by multiple mode

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A Folder Contents

Name	Description
<i>human_gene_association-web.goa.human</i>	human association file
<i>yeast_gene_association-web.sgd</i>	yeast association file
<i>mouse_gene_association-web.mgi</i>	mouse association file

Table 3: *associations* folder content

Name	Description
<i>go-basic.obo</i>	basic terms file
<i>go-web.obo</i>	complete terms file

Table 4: *terms* folder content

Name	Description
<i>yeast_genes.txt</i>	yeast gene tags
<i>yeast_samples.txt</i>	yeast sample tags
<i>yeast_times.txt</i>	yeast time tags
<i>yeast_t0.csv</i>	yeast values at t0
<i>yeast_t1.csv</i>	yeast values at t1
<i>yeast_t2.csv</i>	yeast values at t2
<i>yeast_t3.csv</i>	yeast values at t3
<i>yeast_t4.csv</i>	yeast values at t4
<i>yeast_t5.csv</i>	yeast values at t5
<i>yeast_t6.csv</i>	yeast values at t6
<i>yeast_t7.csv</i>	yeast values at t7
<i>yeast_t8.csv</i>	yeast values at t8
<i>yeast_t9.csv</i>	yeast values at t9
<i>yeast_t10.csv</i>	yeast values at t10
<i>yeast_t11.csv</i>	yeast values at t11
<i>yeast_t12.csv</i>	yeast values at t12
<i>yeast_t13.csv</i>	yeast values at t13

Table 5: *0001* folder content

Name	Description
<i>elutriation_genes.txt</i>	elutriation gene tags
<i>elutriation_samples.txt</i>	elutriation sample tags
<i>elutriation_times.txt</i>	elutriation time tags
<i>elutriation_t00.csv</i>	elutriation values at t00
<i>elutriation_t30.csv</i>	elutriation values at t30
<i>elutriation_t60.csv</i>	elutriation values at t60
<i>elutriation_t90.csv</i>	elutriation values at t90
<i>elutriation_t120.csv</i>	elutriation values at t120
<i>elutriation_t150.csv</i>	elutriation values at t150
<i>elutriation_t180.csv</i>	elutriation values at t180
<i>elutriation_t210.csv</i>	elutriation values at t210
<i>elutriation_t240.csv</i>	elutriation values at t240
<i>elutriation_t270.csv</i>	elutriation values at t270
<i>elutriation_t300.csv</i>	elutriation values at t300
<i>elutriation_t330.csv</i>	elutriation values at t330
<i>elutriation_t360.csv</i>	elutriation values at t360
<i>elutriation_t390.csv</i>	elutriation values at t390

Table 6: 0002 folder content

Name	Description
<i>gds4510_genes.txt</i>	gds4510 gene tags
<i>gds4510_samples.txt</i>	gds4510 sample tags
<i>gds4510_times.txt</i>	gds4510 time tags
<i>gds4510_t2.csv</i>	gds4510 values at t2
<i>gds4510_t4.csv</i>	gds4510 values at t4
<i>gds4510_t6.csv</i>	gds4510 values at t6
<i>gds4510_t8.csv</i>	gds4510 values at t8

Table 7: 0003 folder content

Name	Description
<i>gds4472_genes.txt</i>	gds4472 gene tags
<i>gds4472_samples.txt</i>	gds4472 sample tags
<i>gds4472_times.txt</i>	gds4472 time tags
<i>gds4472_t0.csv</i>	gds4472 values at t0
<i>gds4472_t8.csv</i>	gds4472 values at t8
<i>gds4472_t16.csv</i>	gds4472 values at t16
<i>gds4472_t24.csv</i>	gds4472 values at t24
<i>gds4472_t48.csv</i>	gds4472 values at t48
<i>gds4472_t96.csv</i>	gds4472 values at t96

Table 8: 0004 folder content

Name	Description
<i>gds4262_genes.txt</i>	gds4262 gene tags
<i>gds4262_samples.txt</i>	gds4262 sample tags
<i>gds4262_times.txt</i>	gds4262 time tags
<i>gds4262_t2.csv</i>	gds4262 values at t2
<i>gds4262_t6.csv</i>	gds4262 values at t6
<i>gds4262_t12.csv</i>	gds4262 values at t12
<i>gds4262_t24.csv</i>	gds4262 values at t24

Table 9: 0005 folder content

Name	Description
<i>gds4442_genes.txt</i>	gds4442 gene tags
<i>gds4442_samples.txt</i>	gds4442 sample tags
<i>gds4442_times.txt</i>	gds4442 time tags
<i>gds4442_t12.csv</i>	gds4442 values at t12
<i>gds4442_t24.csv</i>	gds4442 values at t24
<i>gds4442_t48.csv</i>	gds4442 values at t48

Table 10: 0006 folder content

Name	Description
<i>gse4324_genes.txt</i>	gse4324 gene tags
<i>gse4324_samples.txt</i>	gse4324 sample tags
<i>gse4324_times.txt</i>	gse4324 time tags
<i>gse4324_0d.csv</i>	gse4324 values at 0d
<i>gse4324_3d.csv</i>	gse4324 values at 3d
<i>gse4324_7d.csv</i>	gse4324 values at 7d
<i>gse4324_14d.csv</i>	gse4324 values at 14d

Table 11: 0007 folder content

Name	Description
<i>earthquake_genes.txt</i>	earthquake gene tags
<i>earthquake_samples.txt</i>	earthquake sample tags
<i>earthquake_times.txt</i>	earthquake time tags
<i>earthquake_counter.csv</i>	earthquake values at counter
<i>earthquake_35.csv</i>	earthquake values at 35
<i>earthquake_40.csv</i>	earthquake values at 40
<i>earthquake_45.csv</i>	earthquake values at 45
<i>earthquake_50.csv</i>	earthquake values at 50
<i>earthquake_max.csv</i>	earthquake values at max
<i>earthquake_lat.csv</i>	earthquake values at lat
<i>earthquake_lon.csv</i>	earthquake values at lon

Table 12: 0008 folder content

Name	Description
<i>synth01_target.csv</i>	synth01 targets
<i>synth01_t0.csv</i>	synth01 values at t0
<i>synth01_t1.csv</i>	synth01 values at t1
<i>synth01_t2.csv</i>	synth01 values at t2
<i>synth01_t3.csv</i>	synth01 values at t3
<i>synth01_t4.csv</i>	synth01 values at t4
<i>synth01_t5.csv</i>	synth01 values at t5
<i>synth01_t6.csv</i>	synth01 values at t6
<i>synth01_t7.csv</i>	synth01 values at t7
<i>synth01_t8.csv</i>	synth01 values at t8
<i>synth01_t9.csv</i>	synth01 values at t9
<i>synth01_t10.csv</i>	synth01 values at t10
<i>synth01_t11.csv</i>	synth01 values at t11
<i>synth01_t12.csv</i>	synth01 values at t12
<i>synth01_t13.csv</i>	synth01 values at t13
<i>synth01_t14.csv</i>	synth01 values at t14
<i>synth01_t15.csv</i>	synth01 values at t15
<i>synth01_t16.csv</i>	synth01 values at t16
<i>synth01_t17.csv</i>	synth01 values at t17
<i>synth01_t18.csv</i>	synth01 values at t18
<i>synth01_t19.csv</i>	synth01 values at t19

Table 13: 0009 folder content

Name	Description
<i>synth02_target.csv</i>	synth02 targets
<i>synth02_t0.csv</i>	synth02 values at t0
<i>synth02_t1.csv</i>	synth02 values at t1
<i>synth02_t2.csv</i>	synth02 values at t2
<i>synth02_t3.csv</i>	synth02 values at t3
<i>synth02_t4.csv</i>	synth02 values at t4
<i>synth02_t5.csv</i>	synth02 values at t5
<i>synth02_t6.csv</i>	synth02 values at t6
<i>synth02_t7.csv</i>	synth02 values at t7
<i>synth02_t8.csv</i>	synth02 values at t8
<i>synth02_t9.csv</i>	synth02 values at t9
<i>synth02_t10.csv</i>	synth02 values at t10
<i>synth02_t11.csv</i>	synth02 values at t11
<i>synth02_t12.csv</i>	synth02 values at t12
<i>synth02_t13.csv</i>	synth02 values at t13

Table 14: 0010 folder content

Name	Description
<i>synth03_target.csv</i>	synth03 targets
<i>synth03_t0.csv</i>	synth03 values at t0
<i>synth03_t1.csv</i>	synth03 values at t1
<i>synth03_t2.csv</i>	synth03 values at t2
<i>synth03_t3.csv</i>	synth03 values at t3
<i>synth03_t4.csv</i>	synth03 values at t4
<i>synth03_t5.csv</i>	synth03 values at t5
<i>synth03_t6.csv</i>	synth03 values at t6
<i>synth03_t7.csv</i>	synth03 values at t7
<i>synth03_t8.csv</i>	synth03 values at t8
<i>synth03_t9.csv</i>	synth03 values at t9

Table 15: 0011 folder content

Name	Description
<i>synth04_target.csv</i>	synth04 targets
<i>synth04_t0.csv</i>	synth04 values at t0
<i>synth04_t1.csv</i>	synth04 values at t1
<i>synth04_t2.csv</i>	synth04 values at t2
<i>synth04_t3.csv</i>	synth04 values at t3
<i>synth04_t4.csv</i>	synth04 values at t4
<i>synth04_t5.csv</i>	synth04 values at t5
<i>synth04_t6.csv</i>	synth04 values at t6
<i>synth04_t7.csv</i>	synth04 values at t7
<i>synth04_t8.csv</i>	synth04 values at t8
<i>synth04_t9.csv</i>	synth04 values at t9

Table 16: 0012 folder content

Name	Description
<i>synth05_target.csv</i>	synth05 targets
<i>synth05_t0.csv</i>	synth05 values at t0
<i>synth05_t1.csv</i>	synth05 values at t1
<i>synth05_t2.csv</i>	synth05 values at t2
<i>synth05_t3.csv</i>	synth05 values at t3
<i>synth05_t4.csv</i>	synth05 values at t4
<i>synth05_t5.csv</i>	synth05 values at t5
<i>synth05_t6.csv</i>	synth05 values at t6
<i>synth05_t7.csv</i>	synth05 values at t7
<i>synth05_t8.csv</i>	synth05 values at t8
<i>synth05_t9.csv</i>	synth05 values at t9

Table 17: 0013 folder content

Name	Description
<i>toy01_target.csv</i>	toy01 targets
<i>toy01_t0.csv</i>	toy01 values at t0
<i>toy01_t1.csv</i>	toy01 values at t1
<i>toy01_t2.csv</i>	toy01 values at t2
<i>toy01_t3.csv</i>	toy01 values at t3
<i>toy01_t4.csv</i>	toy01 values at t4
<i>toy01_t5.csv</i>	toy01 values at t5
<i>toy01_t6.csv</i>	toy01 values at t6
<i>toy01_t7.csv</i>	toy01 values at t7
<i>toy01_t8.csv</i>	toy01 values at t8
<i>toy01_t9.csv</i>	toy01 values at t9
<i>toy01_t10.csv</i>	toy01 values at t10
<i>toy01_t11.csv</i>	toy01 values at t11
<i>toy01_t12.csv</i>	toy01 values at t12
<i>toy01_t13.csv</i>	toy01 values at t13

Table 18: *0014* folder content

B Changelog

C Future developments

- GUI for TriGenApp.
- GUI for LabApp.
- An automatized process in order to include a new dataset.
- Improvement of the dataset storage.
- TrLab web site in order to launch experiments remotely.
- An automatized process in order to install the application.
- Use another GO analysis tools like Panther (<http://www.pantherdb.org>).